

Lecture 9 Worksheet

Math 124, Fall 2026 at the University of Michigan

Some lectures will have an accompanying worksheet. We will do our best to include these problems directly in the notes as activities or worked examples.

Activity 1: Orthogonal projections in $\mathbb{R}^3$

Let

$$\vec{v} = \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$$

and let P be the plane

$$2x + y - 2z = 0.$$

Find the orthogonal projection of $\vec{v}$ onto P .

- a) Find a normal vector $\vec{n}$ to P .

- b) Find the projection of $\vec{v}$ onto $\ell = \text{span}(\vec{n})$.

- c) Use your answer to find $\text{proj}_P(\vec{v})$.

- d) Verify that your answer lies in P .

Activity 2: Coordinates in an Orthonormal Basis

Consider the vectors

$$\vec{u}_1 = \begin{bmatrix} \frac{3}{7} \\ -\frac{2}{7} \\ \frac{6}{7} \end{bmatrix}, \quad \vec{u}_2 = \begin{bmatrix} \frac{6}{7} \\ \frac{3}{7} \\ -\frac{2}{7} \end{bmatrix}, \quad \vec{u}_3 = \begin{bmatrix} -\frac{2}{7} \\ \frac{6}{7} \\ \frac{3}{7} \end{bmatrix}.$$

- a) Verify that each vector has length 1.

- b) Verify that the vectors are pairwise orthogonal by computing

$$\vec{u}_1 \cdot \vec{u}_2, \quad \vec{u}_1 \cdot \vec{u}_3, \quad \vec{u}_2 \cdot \vec{u}_3.$$

- c) Explain why $(\vec{u}_1, \vec{u}_2, \vec{u}_3)$ is an orthonormal basis of $\mathbb{R}^3$.

$$\vec{u}_1 = \begin{bmatrix} \frac{3}{7} \\ -\frac{2}{7} \\ \frac{6}{7} \end{bmatrix}, \quad \vec{u}_2 = \begin{bmatrix} \frac{6}{7} \\ \frac{3}{7} \\ -\frac{2}{7} \end{bmatrix}, \quad \vec{u}_3 = \begin{bmatrix} -\frac{2}{7} \\ \frac{6}{7} \\ \frac{3}{7} \end{bmatrix}.$$

d) Let

$$\vec{w} = \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix}.$$

Compute

$$\vec{w} \cdot \vec{u}_1, \quad \vec{w} \cdot \vec{u}_2, \quad \vec{w} \cdot \vec{u}_3.$$

e) Use these dot products to write $\vec{w}$ as a linear combination of $\vec{u}_1, \vec{u}_2, \vec{u}_3$.

f) Check your answer by carrying out the linear combination in standard coordinates.

Activity 3: Engineering Application: Forces on a Drone

A drone's orientation is described by three local directions:

$$\vec{u}_1 = \begin{bmatrix} \frac{3}{7} \\ -\frac{2}{7} \\ \frac{6}{7} \end{bmatrix}, \quad \vec{u}_2 = \begin{bmatrix} \frac{6}{7} \\ \frac{3}{7} \\ -\frac{2}{7} \end{bmatrix}, \quad \vec{u}_3 = \begin{bmatrix} -\frac{2}{7} \\ \frac{6}{7} \\ \frac{3}{7} \end{bmatrix}.$$

These vectors point in the drone's forward, rightward, and upward directions, respectively. They form an orthonormal basis of $\mathbb{R}^3$.

A force acting on the drone has global coordinates

$$\vec{F} = \begin{bmatrix} 4 \\ -1 \\ 5 \end{bmatrix} \text{ N.}$$

- a) Find the component of $\vec{F}$ in the drone's forward direction.

$$\vec{F} \cdot \vec{u}_1 = \underline{\hspace{2cm}}$$

- b) Find the component of $\vec{F}$ in the drone's rightward direction.

$$\vec{F} \cdot \vec{u}_2 = \underline{\hspace{2cm}}$$

- c) Find the component of $\vec{F}$ in the drone's upward direction.

$$\vec{F} \cdot \vec{u}_3 = \underline{\hspace{2cm}}$$

- d) Use your answers to write $\vec{F}$ as a linear combination of the drone's local directions:

$$\vec{F} = \underline{\hspace{1cm}} \vec{u}_1 + \underline{\hspace{1cm}} \vec{u}_2 + \underline{\hspace{1cm}} \vec{u}_3.$$

- e) Check your answer by carrying out the linear combination in standard coordinates.

- f) Interpret the signs and relative sizes of the three components. In which local direction does most of the force act?